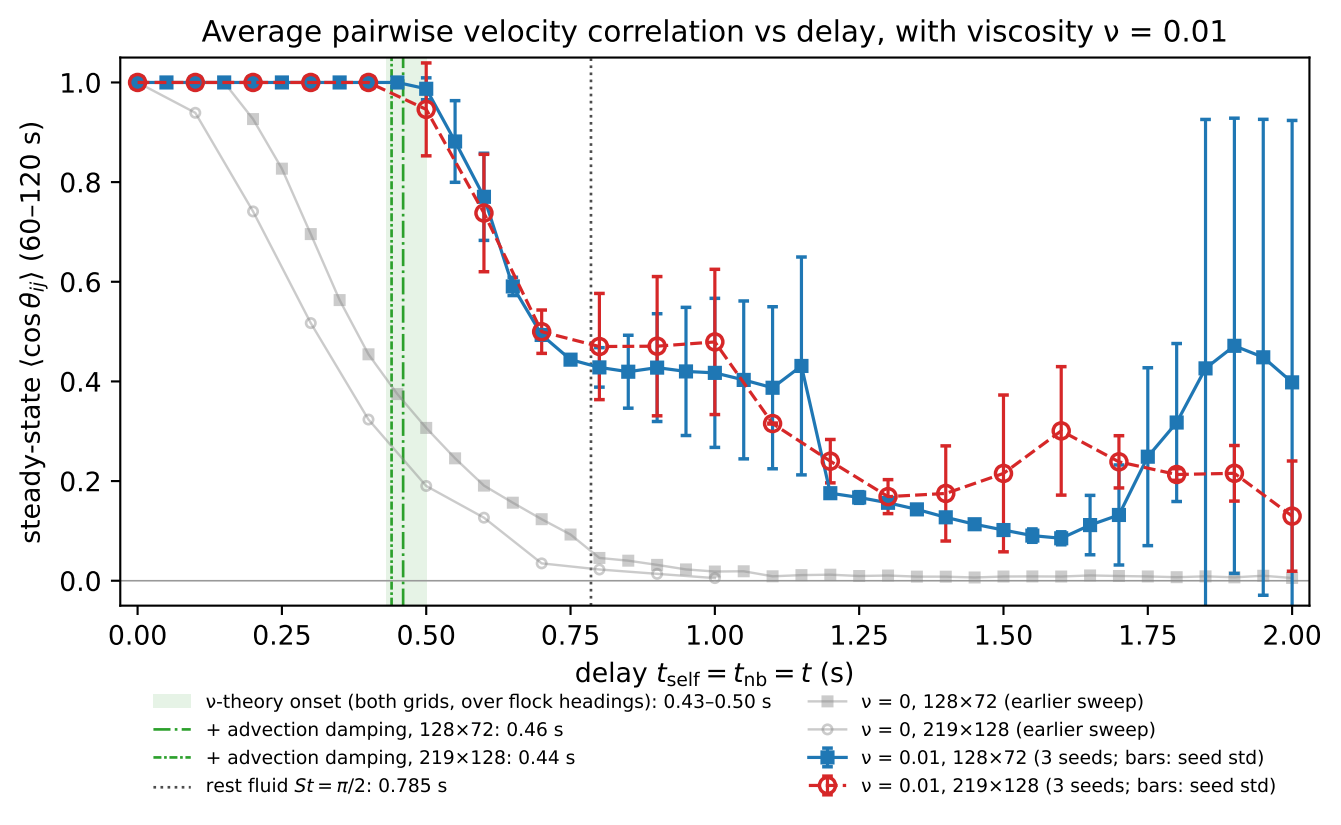
Average pairwise velocity correlation vs delay, with viscosity ν = 0.01
steady-state ⟨cos θ_ij⟩ (60–120 s)
1.0
0.8
0.6
0.4
0.2
0.0
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
delay t_self = t_nb = t (s)
ν-theory onset (both grids, over flock headings): 0.43–0.50 s
+ advection damping, 128×72: 0.46 s
+ advection damping, 219×128: 0.44 s
rest fluid St = π/2: 0.785 s
ν = 0, 128×72 (earlier sweep)
ν = 0, 219×128 (earlier sweep)
ν = 0.01, 128×72 (3 seeds; bars: seed std)
ν = 0.01, 219×128 (3 seeds; bars: seed std)